Before Convergence: A Formal Abstract-Art Interpretation of Deep Neural Network Loss Trajectories

Compile Error and Segmentation Fault

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Objectives

This paper aims to:

- To publicize the seemingly miraculous trajectories of Loss curves.
- To release the frustration accumulated during unsuccessful model training.
- To establish a formal correspondence between optimization dynamics and abstract artistic structures.

Introduction

The training process of deep neural networks is typically characterized by the Loss curve, whose dynamic changes reflect the optimization path and convergence state in the parameter space. Traditional studies mainly analyze the Loss curve from the perspectives of numerical optimization and generalization ability, while paying less attention to the visual features and structural tension presented by its formal structure. From a formalist perspective of abstract art, this paper analyzes the morphological characteristics of the Loss trajectory during the training process of deep neural networks. We point out that, prior to convergence, the Loss curve exhibits typical abstract formal features such as fragmentation, nonlinear oscillation, structural rupture, and staged reorganization. Through segmented analysis of trajectories at different training stages, this paper constructs a "formal mapping framework" that translates phenomena such as oscillations, plateaus, and abrupt changes in optimization dynamics into compositional elements in abstract art. The experimental analysis is based on typical training curve cases, verifying the explanatory power of this formal perspective for understanding the dynamic structure of Loss. This paper aims to provide an interdisciplinary formal analysis method, offering new visual and structural metaphors for understanding the training process of deep learning.

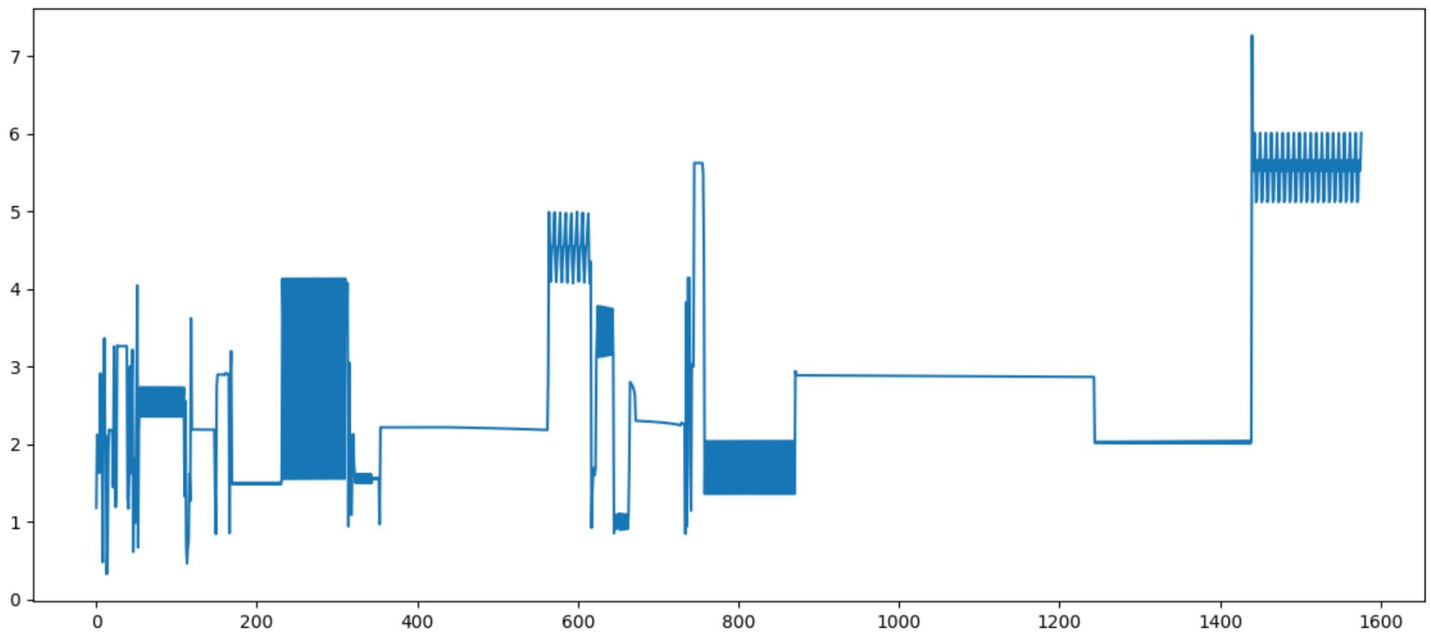

Figure 1: A Loss curve generated during an actual training process, with the horizontal axis representing training epochs and the vertical axis representing the Loss value.

Figure 2: The classic abstract painting Guernica.

Methods

This paper focuses solely on its formal characteristics and compares them with the structural dynamics of the Loss trajectory. Adopting a formalist analytical perspective, we use fragmented composition, nonlinear structure, and tension-generation mechanisms in abstract art as a reference framework to conduct structural decomposition and mapping analysis of the staged forms of the Loss curve during the training process of deep neural networks.

Related Work

This paper is consistent in thought with previous research in the field of biology[1, 2].

Conclusion

This paper analyzes the structural forms of the Loss trajectory during the training process of deep neural networks from a formalist perspective. Through a case study, we point out that, prior to convergence, the Loss curve exhibits characteristics such as fragmented segmentation, multidirectional tension, and structural reorganization, and that its mode of morphological organization shows significant similarity to the formal logic of abstract painting.

References

[1] Feichong Xiao and Guier Ding. Gel's thousand-mile vision: An accidental emergence of a thousand miles of rivers and mountains in a polyacrylamide medium and its scientific–aesthetic implications. *Rubbish*, 2026.

[2] Anonymous. My western blot looks like a panda. *Rubbish*, 2026.

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